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To cite this article: G Suryanta *et al* 2024 *IOP Conf. Ser.: Earth Environ. Sci.* **1352** 012071

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Spatial analysis of land cover changes in mangrove and peatland forests in North Sumatra, Indonesia

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Abstract. Forest management in Indonesia can be observed through land cover change as a macro indicator for forest loss or gains. In order to have a clear understanding of the trend of the land cover change in peatland and mangroves, we conduct the land cover change analysis using the last decadal or 10-year period. The land typologies used as analysis units are peat and mangrove ecosystem. The research area is located in North Sumatra Province, occupying a peat area of 50,089.9 ha and mangrove area of 86,907 ha. The spatial analysis includes land cover maps overlay and classification using GIS. The 2012-2022 analysis results indicate a trend of an increase in peat reforestation by 819.6 ha and deforestation by 5,043.5 ha, while the trend of mangrove reforestation increased by 12,801.5 ha and deforestation by 4,669.5 ha. Land cover changes in the peatland are predominantly derived due to conversion of plantation to secondary swamp forest and conversion of secondary swamp forest to bare land. In the mangrove ecosystem land cover changes are mainly derived due to the regrowth of shrubs to secondary mangrove forests and due to conversion or degradation of secondary mangrove forests to shrubs.

1. Introduction

Peatland and mangroves are two important wetlands that contribute significantly to climate mitigation and adaptation globally. Indonesia owing 3.3 million ha of mangroves [1] consists of about 23% of the world's mangrove forests. With the vast extent and high biodiversity of Indonesian mangroves, mangrove loss still occurred with net deforestation rate of 12,818 ha year⁻¹ in 2009-2019 [2]. Peatland in Indonesia is estimated at 13.43 million ha (30.5% of the world's tropical peatland), which makes Indonesia as the country with the largest tropical peatland in the world [3]. However, peatland management in Indonesia is challenged by peatland fires due to land use change, both inside and outside forest areas [4]. Considering the challenges that need to be solved on wetlands management, the Government of Indonesia had mainstreamed wetlands management, especially peat and mangroves, into the national development plans. Since the last decade Indonesia had decreed Government's regulations



as outlined in Government regulations on peat conversion moratorium (Presidential Regulation No 5/2019) and accelerated mangrove rehabilitation by establishing Peatland and Mangrove Restoration Agency (Presidential Regulation No. 120/2020). One of the goals of these regulations are to meet the climate targets and to halt global warming due to rising temperatures from anthropogenic activities.

To halt mangrove deforestation, accelerating mangrove rehabilitation has been carried out of around 1,000 ha/year. Since 2020, the acceleration of mangrove rehabilitation has been conducted until 2025 with a target of hundreds of hectares throughout Indonesia's mangroves [5]. Until 2022, the Ministry of Environment and Forestry and BRGM have worked on an area of mangrove rehabilitation 87,763 ha, both inside forest areas and outside forest areas, with the funding schemes from the national funds, private companies, international donors, and others. Peatland moratorium that had been implemented since 2019 resulted in significant decrease of peatland conversion and the occurrence of peatland fires [6]. Peatland and wetlands in North Sumatra are experiencing a significant decrease/increase in the last decade [7-8]. In order to provide data activities of wetlands in North Sumatra, this study aimed to analyze the current trend of peatland and mangrove cover change in North Sumatra. This information will be needed to estimate the emission baseline of peatland and mangroves at the provincial level.

2. Methods

This study used Geographic Information System (GIS) to analyze land cover change of peatland and mangrove forests from 2012 to 2022. Land cover maps were produced by the Ministry of Environment and Forestry of Indonesia, and had been overlaid with the Peat Map shapefile data from the 1984 RePPProT map and the 2021 National Mangrove Map.

Reforestation is replanting activity in degraded areas and is part of rehabilitation and restoration activities. Deforestation is defined as the loss of forest areas due to anthropogenic activities or natural disturbances. This study classified any land cover changes in peatland and mangrove forests and their trajectories whether the change will result in carbon emissions (-) or sequestration (+) (Table 1).

Table 1. Classification of reforestation and deforestation for peatland and mangrove forests

	Primary dryland forest	Secondary dryland forest	Primary mangrove forest	Secondary mangrove forest	Primary swamp forest	Secondary swamp forest	Plantation forest
a f t e r	Primary dryland forest	Remaining forest - P (+)	Remaining forest - GRW (+)	Remaining forest - P (-)	Remaining forest - Deg (-)	Remaining forest - P (+)	Remaining forest - GRW (+)
	Secondary dryland forest	Remaining forest - Deg (-)	Remaining forest - S (+)	Remaining forest - Deg (-)	Remaining forest - S (-)	Remaining forest - Deg (-)	Remaining forest - S (+)
	Primary mangrove forest	Remaining forest - P (+)	Remaining forest - GRW (+)	Remaining forest - P (+)	Remaining forest - GRW (+)	Remaining forest - P (+)	Remaining forest - GRW (+)
	Secondary mangrove forest	Remaining forest - GRW (+)	Remaining forest - S (+)	Remaining forest - Deg (-)	Remaining forest - S (+)	Remaining forest - Deg (-)	Remaining forest - S (+)
	Primary swamp forest	Remaining forest - P (+)	Remaining forest - GRW (+)	Remaining forest - P (+)	Remaining forest - GRW (+)	Remaining forest - P (+)	Remaining forest - GRW (+)
t h a n	Secondary swamp forest	Remaining forest - Deg (-)	Remaining forest - S (+)	Remaining forest - Deg (-)	Remaining forest - S (+)	Remaining forest - Deg (-)	Remaining forest - S (+)
	Plantation forest	Remaining forest - Deg (-)	Remaining forest - Deg (-)	Remaining forest - Deg (-)	Remaining forest - Deg (-)	Remaining forest - Deg (-)	Remaining forest - Deg (-)
	Shrubs	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Swamp shrubs	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Dryland agriculture	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
2 0 1 2	Mixed dryland agriculture	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Paddy field	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Ponds	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Settlement-trans	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Cultivation field	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
1 2	Settlement	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Airport/Port	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Bare land	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Mining field	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Water body	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
1 2	Swamp	Degradation - S (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Savanna	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Clouds	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted

		before (2012)							
		Shrubs	Swamp shrubs	Dryland agriculture	Mixed dryland agriculture	Paddy field	Ponds	Settlements-trans	Cultivation field
a	Primary dryland forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Secondary dryland forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
f	Primary mangrove forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Secondary mangrove forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
e	Primary swamp forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Secondary swamp forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
r	Plantation forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Shrubs	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Deforestation (-)
t	Swamp shrubs	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Deforestation (-)
	Dryland agriculture	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Deforestation (-)
h	Mixed dryland agriculture	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Deforestation (-)
	Paddy field	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Deforestation (-)
a	Ponds	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)	Deforestation (-)
	Settlement-trans	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Deforestation (-)	Deforestation (-)	Remaining NF - GRW (-)	Deforestation (-)
n	Cultivation field	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
	Settlement	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
2	Airport/Port	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
	Bare land	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
0	Mining field	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
	Water body	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
1	Swamp	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
	Savanna	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
2	Clouds	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted

		before (2012)							
		Settlement	Airport/port	Bare land	Mining field	Water body	Swamp	Savanna	Clouds
a	Primary dryland forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - P (+)
	Secondary dryland forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - S (+)
f	Primary mangrove forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - P (+)
	Secondary mangrove forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - S (+)
e	Primary swamp forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - P (+)
	Secondary swamp forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Remaining forest - S (+)
r	Plantation forest	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Shrubs	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
t	Swamp shrubs	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Dryland agriculture	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
h	Mixed dryland agriculture	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)	Reforestation (+)
	Paddy field	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
a	Ponds	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
	Settlement-trans	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
n	Cultivation field	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
	Settlement	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
2	Airport/Port	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
	Bare land	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
0	Mining field	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - GRW (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
	Water body	Deforestation (-)	Deforestation (-)	Deforestation (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)	Remaining NF - Deg (-)
1	Swamp	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
	Savanna	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)	Remaining NF - GRW (+)
2	Clouds	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted	Not counted

Remarks: green color shows the changes of land cover due to reforestation while orange color shows the land cover changes due to deforestation.

3. Results and Discussion

The annual land cover change for each land cover types are as shown in Figure 1.

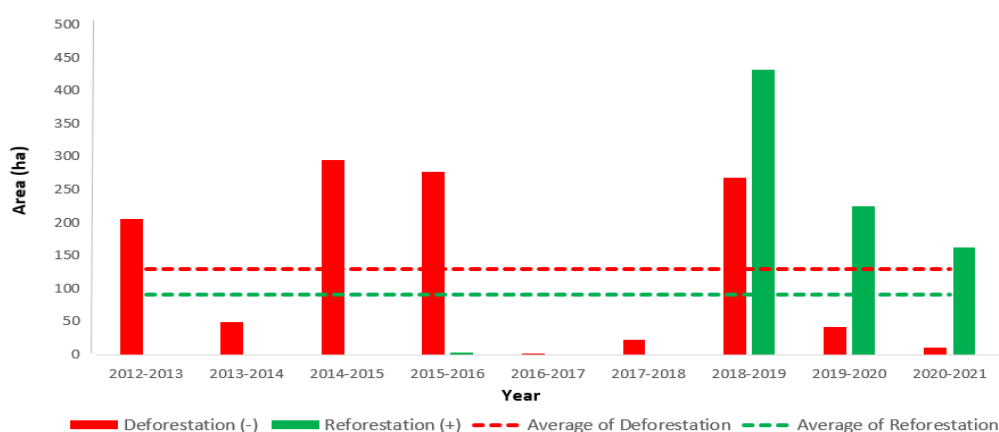


Figure 1. Reforested and deforested areas in the peatlands of North Sumatra from 2012-2022

Peatland deforestation from 2012 to 2015 was quite high covering an area of 49.32 ha to 293.96 ha, while a relatively small deforestation rate from 2016 to 2018 (0.20-21.81 ha) was observed (Figure 1). Deforestation in 2019 ranged between 10.8-267.6 ha. Peat reforestation reached 430.9 ha in 2019, and decreased to 224.5 ha and 161.56 ha in 2020–2021 respectively. The mean of peatland deforested area in North Sumatra from 2012-2021 is 129,62 ha or a deforestation rate of 129,62 ha/year.

The cumulative area of peatland reforestation in North Sumatra from 2012 to 2022 is 819.42 ha which was dominated by the change of plantation into secondary swamp forest (Figure 2). The cumulative area of deforestation on peatland in North Sumatra from 2012 to 2022 was 1,166.54 ha which was dominated by the change of secondary swamp forest to open land dominates (Figure 3).

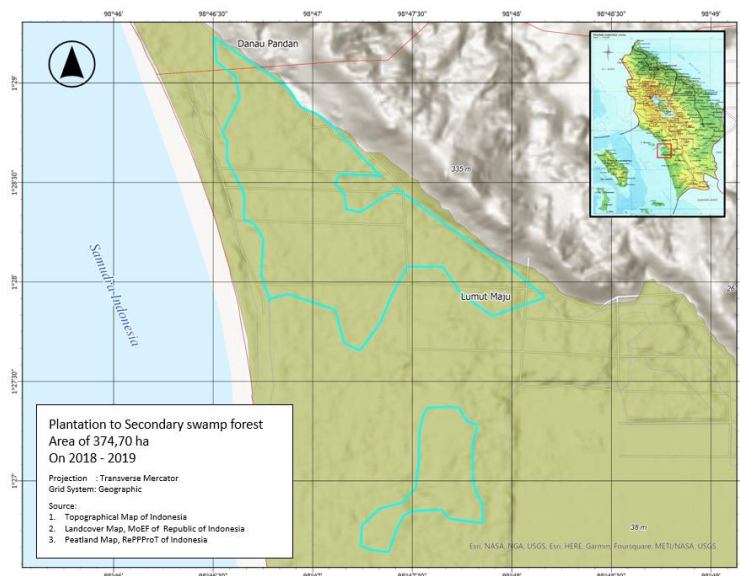


Figure 2. Peatland cover change from plantation to secondary swamp forest due to reforestation activities covering an area of 374.70 ha in North Sumatra.

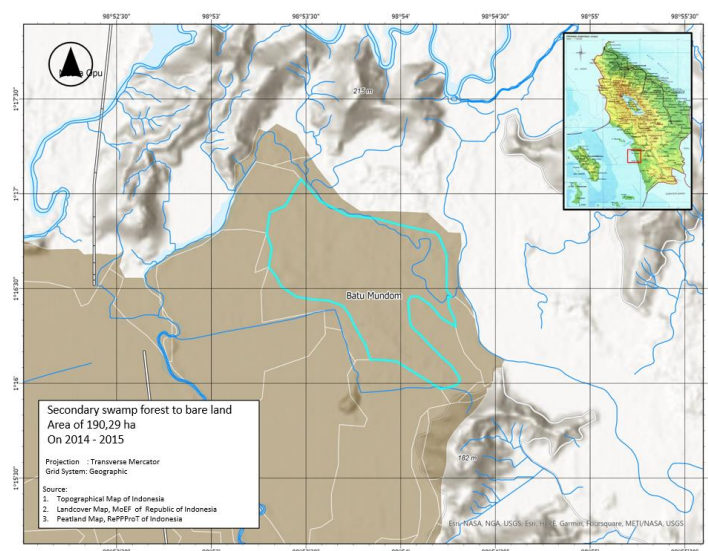


Figure 3. Peatland changes from secondary swamp forest to open land due to deforestation in North Sumatra.

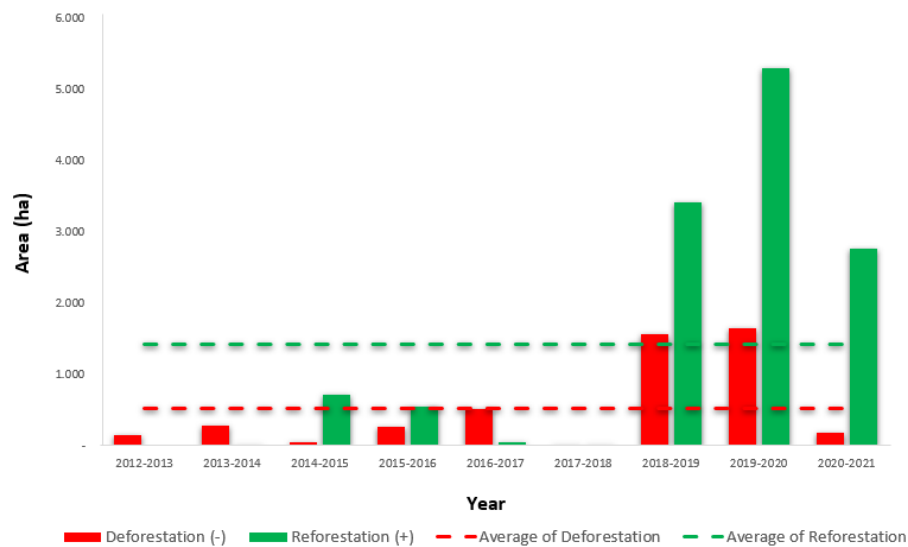


Figure 4. Deforestation and reforestation activities in mangrove forests of North Sumatra during 2012-2022

In the period 2012 to 2019 there was an increase in mangrove deforestation from 154.3 ha to 1,640.39 ha, then decreased to 174.21 ha in 2020. Likewise, the trend of mangrove reforestation tends to increase from 2014 to 2019, namely 713.38 ha to 5,295.55 ha; whereas in 2020 there is a decrease of 2,760.63 ha (Figure 4). The cumulative area of mangrove reforestation in North Sumatra from 2012 to 2022 was 12,797.41 ha which was due to the growth of swamp shrubs into secondary mangrove forest [9-10]. The mean of reforested mangrove area in North Sumatra from 2012-2022 was 1,422.40 ha or reforestation rate of 1,422.40 ha/year (Figure 5).

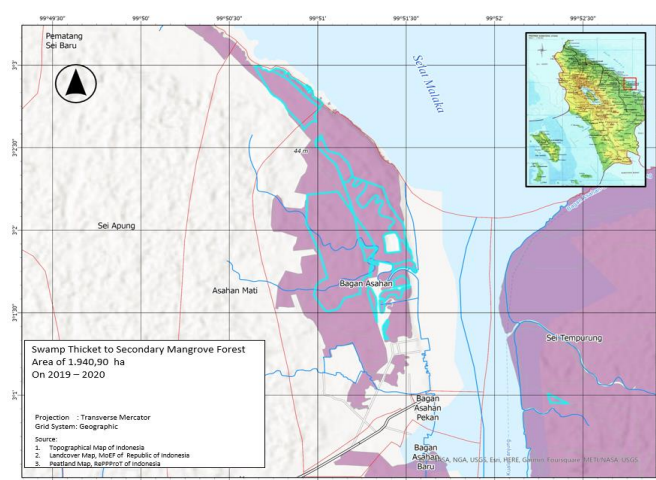


Figure 5. Land cover change from shrub swamp forest to secondary mangrove forest due to reforestation activities covering an area of 1,940.9 ha in North Sumatra.

The cumulative area of mangrove deforestation in North Sumatra from 2012 to 2022 was 4,668.96 ha which was predominantly due to the change of secondary mangrove forest to swamp shrubs (Figure 6). The mean of mangrove deforested area from 2012 to 2022 in North Sumatra is 27.793,6 ha or a deforestation rate of 2779 ha/year.

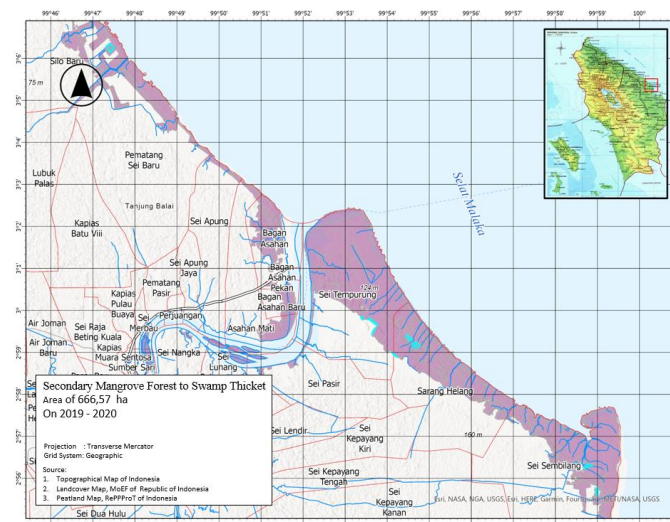


Figure 6. Land cover change from secondary mangrove forest to swamp shrubs due to deforestation covering an area of 666.57 ha in North Sumatra.

3. Conclusion

In 2012 to 2022 the rate of reforestation and deforestation on peatlands in North Sumatra (both inside and outside forest areas), covered an area of around 819.62 ha and 1,166.54 ha respectively. The drivers of peatland reforestations were due to the change of plantations into secondary swamp forest and peatland deforestation was mainly due to secondary swamp forest that was converted into open land/bare land. Likewise, mangrove reforestation in North Sumatra covers an area of around 48.709,8 ha and a reforestation rate of around 4871 ha/year. Mangrove reforestation is predominantly due to the growth of shrubs into secondary mangrove forests while mangrove deforestation is mainly due to the change of secondary mangrove forests to shrubs. On a macro scale, it can be compared that the rate of reforestation on peatlands over the past 10 years has been higher than the rate of deforestation. Likewise, the rate of reforestation in mangroves over the past 10 years has been higher than the rate of deforestation. More in-depth study through further research and more detailed land cover change mapping using high resolution imageries and UAVs should be conducted to support acceleration of mangrove and peat restoration in Indonesia.

Acknowledgments

This work was supported by Basic Research Collaboration Scheme 2022-2024 (e-Asia Joint Research Program) (079/UN5.2.3.1/PPM/KP-DRTPM/L/2022) from the Ministry of Education, Culture, Research and Technology of Government of Indonesia. This study was also partly supported by the Joint Collaboration Riset Rumah Program Organisasi Riset Hayati dan Lingkungan 2023, National Research and Innovation Agency. Center of Excellence for Mangrove, Universitas Sumatera Utara covered the study and APC of this paper.

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